Boundary CFT_3-on-S^3 observables on the truncated GeoVac spectral triple: partition function, wedge KMS entropy, and Casini–Huerta–Myers modular Hamiltonian

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Abstract

We exhibit three complementary boundary-side CFT_3-on-S^3 observables on the truncated GeoVac spectral triple, each computed from the same wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ via a structurally distinct operation: *(A)* the universal F-theorem coefficient F_{S^3}, extracted via the spectral-zeta derivative $-\frac{1}{2}\zeta'_\Delta(0)$, reproduces the Klebanov–Pufu–Safdi (2011) continuum value bit-exactly for both the conformally coupled scalar ($F_s = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2}$) and the free Camporesi–Higuchi Dirac ($F_D = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2}$); *(B)* the wedge KMS entropy $S(\rho_W) \sim 2\log n_{\max}$ extracts the wedge boundary dimension $\dim(\partial W) = 2$ as a degeneracy log on the lowest K_α^W shell; *(C)* the framework's modular Hamiltonian $K_\alpha^W = J_{\mathrm{polar}}$ identifies, at finite cutoff, with the continuum Casini–Huerta–Myers hemisphere modular Hamiltonian, with GH-convergence rate inherited from [8, 12]. The scalar/Dirac partition functions decompose orthogonally under the master Mellin engine of [2]: $F_D + 2F_s = \log(2)/2$ isolates the M2 (Seeley–DeWitt) sector, $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ isolates the M3 (half-integer Hurwitz / odd-zeta) sector. This is the first verification of the master Mellin engine on non-GeoVac-internal physics observables. The bulk-side identifications (Ryu–Takayanagi minimum surfaces, JLMS bulk-modular reconstruction, AdS_4 / H^4 dual geometry) are blocked at the framework's existing infrastructure: zero AdS / H^4 machinery, and the Wick-rotation $S^3 \to H^3$ supplies no natural discrete subgroup [5].

1 Introduction

The GeoVac spectral triple [1, 6] is a finite-cutoff operator-system truncation of the Camporesi–Higuchi spectral triple [18] on the round S^3. Its modular structure on the hemispheric wedge has been developed across Papers 42–49 [9–16]: the four-witness Wick-rotation theorem closes at the operator-system level (Riemannian, finite cutoff) via the geometric BW-α generator $K_\alpha^W = J_{\mathrm{polar}}$ with integer spectrum, and via the Tomita–Takesaki BW-γ modular operator on the GNS Hilbert–Schmidt space; the modular flow $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$ closes bit-exactly. The K^+-weak-form Lorentzian propinquity convergence [12] extends the construction to signature $(3, 1)$, with the strong-form extension [13] and the norm-resolvent non-compact-carrier closure [14] completing the Lorentzian arc. The Krein–Mondino–Sämann bridge [15] and the OSLPLS strong-form bridge [16] provide the synthetic-Lorentzian geometric interpretation.

The present paper turns from the math.OA structural content of the modular machinery to its *physics-side observables.* We ask whether the framework's wedge KMS state ρ_W, evaluated on boundary observables corresponding to standard CFT_3-on-S^3 calculations, reproduces the known continuum values, and whether the resulting structural pattern exhibits any genuinely new content beyond a numerical match.

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Headline results. We establish three complementary boundary-side observables and one structural decomposition theorem:

1. **Partition function match** (Sec. 3): the universal F-theorem coefficient extracted from the framework's spectral-zeta machinery agrees *bit-exactly* with the Klebanov–Pufu–Safdi (2011) [21] continuum values for both the conformally coupled scalar and the free Camporesi–Higuchi Dirac on the round S^3.

2. **Master Mellin engine decomposition** (Theorem 3.3): the scalar and Dirac partition functions decompose into linear combinations of $\log 2$ and $\zeta(3)/\pi^2$ that project orthogonally onto the M2 and M3 sectors of the master Mellin engine [2]. This is the first verification of the master Mellin engine on non-GeoVac-internal physics observables.

3. **Wedge KMS entropy structural classification** (Sec. 4): the von Neumann entropy of ρ_W scales as $S(\rho_W) \sim 2\log n_{\max}$, with the slope $2 = \dim(\partial W)$ identifying the wedge boundary dimension as a degeneracy log on the lowest K_α^W shell. The F-coefficient does *not* appear in $S(\rho_W)$, demonstrating that the framework's spectral-zeta side and wedge KMS state side encode *different* continuum observables on the same modular state.

4. **Casini–Huerta–Myers identification** (Sec. 5): the framework's modular Hamiltonian K_α^W identifies, at finite cutoff, with the continuum CHM hemisphere modular Hamiltonian $K_{\text{cont}} = 2\pi \int_{H_+} \xi \cdot T^{00}\, d\Omega$ [25]; convergence $K_\alpha^W \to K_{\text{cont}}$ as $n_{\max} \to \infty$ is bounded by the Paper 38 GH-convergence rate $\Lambda \le C_3 \cdot \gamma_{n_{\max}}$ with $\gamma_{n_{\max}} \sim (4/\pi)\log(n_{\max})/n_{\max}$.

Bulk-side scope. Bulk-side identifications — Ryu–Takayanagi minimum-surface entanglement entropy [27], the JLMS [28] bulk-modular reconstruction, and AdS_4 / H^4 dual geometry — are blocked at the framework's existing infrastructure: zero AdS_4 / H^4 machinery exists, and a previous attempt at the natural Wick continuation $S^3 \to H^3$ [5] found no discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants. We document this scope statement honestly in Sec. 6, with no claim that the bulk side is reachable from the framework's current axiom set.

2 Setup

2.1 The GeoVac spectral triple on S^3

The Fock projection [1, 17] maps the single-particle hydrogen Hamiltonian conformally to the Laplace–Beltrami operator on S^3 via stereographic projection in momentum space. The framework's spectra on the unit S^3 are exact in $\mathbb{Q}$:

- **Scalar Laplacian.** Eigenvalues $\lambda_n^2 = n(n+2)$ with multiplicity $d_n = (n+1)^2$ for $n = 0, 1, 2, \ldots$. After the conformal mass shift $\xi R_{\text{scalar}} = \frac{1}{8} \cdot 6 = \frac{3}{4}$ for the conformally coupled scalar, the spectrum becomes $(n + \frac{1}{2})(n + \frac{3}{2})$ with the same multiplicity.

- **Camporesi–Higuchi Dirac.** Eigenvalues $|\lambda_n| = n + \frac{3}{2}$ with multiplicity $g_n^{\text{Dirac}} = 2(n+1)(n+2)$ for the Weyl (single chirality) sector, $4(n+1)(n+2)$ for the full 4-component Dirac. Both sectors carry the same Hurwitz-zeta structure under analytic continuation [3, 4].

The truncation at level $n_{\max}$ retains all $n \le n_{\max}$ shells and defines the operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}} \mathcal{A} P_{n_{\max}}$ in the Connes–van Suijlekom framework [19]. Convergence $\mathcal{O}_{n_{\max}} \to \mathcal{O}_{S^3}$ in the Latrémolière propinquity holds at rate $\gamma_{n_{\max}} \sim (4/\pi)\log(n_{\max})/n_{\max}$ [8]; the rate constant $4/\pi$ is the M1 Hopf-base-measure signature of the master Mellin engine [2].

2.2 Wedge KMS state

The hemispheric wedge W corresponds, in the truncated full-Dirac basis, to the half-space with $m_j > 0$ along the wedge-fixing axis. Per [9] Def 5.1, the BW-α modular Hamiltonian on the wedge is

$$K_\alpha^W = J_{\mathrm{polar}}$$

where J_{polar} is the rotation generator preserving the wedge, with integer eigenvalues $2m_j \in \{1, 3, 5, \ldots, 2n_{\max}-1\}$ and multiplicity $g(2k+1) = (n_{\max}-k)(n_{\max}-k+1)$ for $k = 0, \ldots, n_{\max}-1$. The wedge KMS state at canonical BW normalization is

$$\rho_W = \frac{e^{-K_\alpha^W}}{Z}, \qquad Z = \mathrm{Tr}_{\mathcal{H}_W}\left(e^{-K_\alpha^W}\right).$$

This is the substrate for all three boundary-side observables that follow.

3 Partition function: bit-exact match to KPS

3.1 Continuum reference

For the round unit S^3, the free energies of free CFTs computed via Hurwitz-zeta regularization are [21]:

$$F_s^{\mathrm{KPS}} = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} \approx 0.063807, \tag{1}$$

$$F_D^{\mathrm{KPS}} = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} \approx 0.218960, \tag{2}$$

for the conformally coupled scalar and the free massless Weyl Dirac, respectively. Both values are expressed in terms of the functional determinant identities $F = -\frac{1}{2}\zeta_\Delta'(0)$ (scalar) and $F = \zeta_{|D|}'(0)$ (Dirac), with the analytic continuation of the divergent spectral sum supplied by the standard zeta-function regularization. The transcendental signature involves only $\log 2$, $\zeta(3)$, and π^2.

3.2 Framework's Hurwitz machinery

The framework's spectral zeta on the truncated S^3 admits a closed-form Hurwitz expansion. For the Dirac case, the Dirichlet series can be written exactly via the half-integer Hurwitz identity $\zeta_H(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$:

$$D_{\mathrm{Dirac}}(s) = \sum_{n=0}^{\infty} \frac{g_n^{\mathrm{Dirac}}}{|\lambda_n|^s} = 2\left(2^{s-2} - 1\right)\zeta_R(s-2) - \tfrac{1}{2}\left(2^s - 1\right)\zeta_R(s). \tag{3}$$

This identity is encoded in the production module `geovac/qed_two_loop.py` and is verified symbolically across integer s.

For the conformally coupled scalar, the spectral zeta admits the Hurwitz-asymptotic expansion

$$\zeta_\Delta(s) = \sum_{n=0}^{\infty} \frac{(n+1)^2}{[(n+1/2)(n+3/2)]^s} = \sum_{k\geq 0} g_k(s)\, \zeta_R(2s + 2k - 2), \tag{4}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ the binomial-like coefficient.

3.3 Bit-exact match: scalar

Differentiating (4) at $s=0$ and using $g_k(0)=\delta_{k,0}$, $g_k'(0)=1/(k\cdot 4^k)$ for $k\geq 1$, plus the standard identities $\zeta_R'(-2)=-\zeta(3)/(4\pi^2)$, $\zeta_R(0)=-\frac{1}{2}$, $\zeta_R(2k-2)$ expressible in $\pi^{2k-2}\cdot\mathbb{Q}$ for $k\geq 2$:

$$\zeta_\Delta'(0)=2\zeta_R'(-2)+\sum_{k\geq 1}\frac{\zeta_R(2k-2)}{k\cdot 4^k}. \tag{5}$$

Theorem 3.1 (Bit-exact scalar partition function)**.** *The framework's scalar zeta-derivative satisfies*

$$F_s:=-\tfrac{1}{2}\,\zeta_\Delta'(0)=\frac{\log 2}{8}-\frac{3\zeta(3)}{16\pi^2}=F_s^{KPS}, \tag{6}$$

in bit-exact agreement with (1).

Proof sketch. The asymptotic Hurwitz expansion (5) converges geometrically in k at rate 1/4 per term; truncation at $k_{\max}=100$ gives 61+ matching digits at dps = 200 when compared to $-2F_s^{\mathrm{KPS}}$ via `mpmath`. PSLQ at coefficient bound 10^6 and tolerance 10^{-40} finds the integer relation $[8,2,-3]$:

$$8\cdot\zeta_\Delta'(0)+2\log 2-3\cdot\frac{\zeta(3)}{\pi^2}=0,$$

which is exactly the claimed identity. See driver `debug/ads_track_a_scalar_partition_function.py` and JSON `debug/data/ads_track_a_scalar_partition_function.json` for full computation. □

3.4 Bit-exact match: Dirac

Differentiating (3) symbolically at $s=0$, using the same Riemann-zeta identities at $s=-2,0$:

$$D_{\mathrm{Dirac}}'(0)=\tfrac{1}{2}\log 2\cdot\zeta_R(-2)+\left(-\tfrac{3}{2}\right)\zeta_R'(-2)-\tfrac{1}{2}\left[\log 2\cdot\zeta_R(0)+0\right] \tag{7}$$

$$=0-\tfrac{3}{2}\cdot\left(-\tfrac{\zeta(3)}{4\pi^2}\right)-\tfrac{1}{2}\cdot\left(-\tfrac{\log 2}{2}\right)=\frac{3\zeta(3)}{8\pi^2}+\frac{\log 2}{4}. \tag{8}$$

Theorem 3.2 (Bit-exact Dirac partition function)**.** *The framework's Dirac zeta-derivative satisfies, as an exact symbolic identity,*

$$F_D:=D_{\mathrm{Dirac}}'(0)=\frac{\log 2}{4}+\frac{3\zeta(3)}{8\pi^2}=F_D^{KPS}, \tag{9}$$

in bit-exact agreement with (2).

Proof. Direct symbolic computation; `sympy.simplify(D'(0) - F_D)` returns 0. See driver `debug/ads_track_a_dirac_partition_function.py`. □

3.5 Master Mellin engine decomposition

The master Mellin engine of [2] §III.7 classifies the transcendental signatures of GeoVac observables via the three sub-mechanisms

$$\mathcal{M}[\mathrm{Tr}(D^k\cdot e^{-tD^2})],\qquad k\in\{0,1,2\},$$

labeled M1 (Hopf-base measure, $\sqrt{\pi}\cdot\mathbb{Q}$ ring at $k=0$), M2 (Seeley–DeWitt heat-kernel, $\sqrt{\pi}\cdot\mathbb{Q}\oplus\pi^2\cdot\mathbb{Q}$ ring at $k=2$), and M3 (half-integer Hurwitz / vertex-parity, $\zeta(3)/\pi^2$ and similar odd-zeta-over-pi ratios at $k=1$).

Theorems 3.1 and 3.2 exhibit F_s and F_D as linear combinations of $\log 2$ (an M2 transcendental, present via $\zeta_R'(0)=-\frac{1}{2}\log(2\pi)$ in the underlying heat-kernel asymptotic) and $\zeta(3)/\pi^2$ (an M3 transcendental, present via $\zeta_R'(-2)=-\zeta(3)/(4\pi^2)$ in the half-integer Hurwitz-zeta derivative).

Theorem 3.3 (Scalar/Dirac M2–M3 orthogonal decomposition)**.** *The free energies F_s and F_D project orthogonally onto the M2 and M3 axes of the master Mellin engine via the dual-basis combinations*

$$F_D + 2\,F_s = \frac{\log 2}{2} \qquad \textit{(isolates M2; all } \zeta(3) \textit{ cancels)}, \tag{10}$$

$$F_D - 2\,F_s = \frac{3\zeta(3)}{4\pi^2} \qquad \textit{(isolates M3; all } \log 2 \textit{ cancels)}. \tag{11}$$

Proof. Substituting (1) and (2):

$$F_D + 2F_s = \left(\tfrac{\log 2}{4} + \tfrac{3\zeta(3)}{8\pi^2}\right) + 2\left(\tfrac{\log 2}{8} - \tfrac{3\zeta(3)}{16\pi^2}\right) = \tfrac{\log 2}{4} + \tfrac{\log 2}{4} = \tfrac{\log 2}{2},$$

and similarly $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$. The two combinations span a 2-dimensional space; the projection coefficients $(1, +2)$ and $(1, -2)$ act as a dual basis for the M2/M3 decomposition on the (scalar, Dirac) plane. □

Remark 3.4 (First verification on non-GeoVac-internal data)**.** Theorem 3.3 is the first verification of the master Mellin engine on observables that are *not* GeoVac-internal. Previous verifications — the case-exhaustion theorem of [6] §VIII, the 208/208 Prediction 1 panel of [7], the modular-residual closed form of Sprint MR-B — all operated on framework-internal quantities. The KPS partition functions (1), (2) are independently-published continuum CFT_3 data, and the framework's spectral-zeta machinery reproduces them with the exact M2/M3 transcendental signature the master Mellin engine predicts. This elevates the engine from a structural classification of GeoVac-internal observables to a predictive classification of CFT physics on the round S^3 at finite cutoff.

4 Wedge KMS entropy: state-side observable

4.1 The framework's wedge KMS entropy

The von Neumann entropy of the wedge KMS state ρ_W at canonical BW normalization $\beta = 2\pi$ is computed directly by diagonalizing K_α^W and applying $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$. Sprint BH-Phase0 (2026-05-22) [30] computed $S(\rho_W)$ at $n_{\max} \in \{2, 3, 4, 5, 6, 7\}$, obtaining the log-linear fit

$$S(\rho_W) \approx 1.94 \cdot \log(n_{\max}) + 0.56, \qquad R^2 = 0.99991, \tag{12}$$

with the slope ≈ 2 identifiable as the wedge boundary dimension $\dim(\partial W) = \dim(S^2_{\text{equator}}) = 2$.

4.2 Mechanism: degeneracy log on the lowest K_α^W shell

The K_α^W spectrum is bounded above by $2n_{\max} - 1$ at finite cutoff. At BW canonical normalization, the per-state weight at $K_\alpha^W = 1$ (the equator shell) is e^{-1}/Z, while at $K_\alpha^W = 3$ it drops to e^{-3}/Z, a factor of $e^{-2} \approx 0.135$ suppression per shell. The partition function is therefore dominated by the equator shell of multiplicity $n_{\text{eq}} = n_{\max}(n_{\max} + 1) \sim n_{\max}^2$:

$$Z \approx n_{\text{eq}} \cdot e^{-1} \approx 0.368\, n_{\max}^2,$$

giving effective probability $p_{\text{eq}} \approx 1/n_{\text{eq}}$ per equator state, and hence

$$S(\rho_W) \approx -n_{\text{eq}} \cdot p_{\text{eq}} \log p_{\text{eq}} = \log n_{\text{eq}} \approx 2 \log n_{\max}.$$

The two asymptotic limits in s-deformation $\rho(s) = e^{-sK_\alpha^W}/Z$, $s \to 0^+$ (maximally mixed on $\mathcal{H}_W$, giving $S \to \log\dim\mathcal{H}_W$) and $s \to \infty$ (maximally mixed on the equator shell, giving $S \to \log n_{\text{eq}}$), are saturated bit-exactly [30].

4.3 Structural decomposition: F is in the spectral-zeta side, not the state side

Proposition 4.1 (F vs $S(\rho_W)$ live in different sectors)**.** *The framework's wedge KMS state ρ_W admits two structurally distinct calculational operations that extract different continuum observables:*

- ***Spectral-zeta side:*** *the analytic continuation $-\frac{1}{2}\zeta'_\Delta(0)$ extracts the universal F-theorem coefficient F_s^{KPS} via Theorem 3.1; similarly for F_D^{KPS} via Theorem 3.2. The transcendental signature is in the master Mellin engine M2/M3 ring $\mathbb{Q} \cdot \log 2 \oplus \mathbb{Q} \cdot \zeta(3)/\pi^2$.*

- ***Wedge KMS state side:*** *the von Neumann entropy $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$ extracts the wedge boundary dimension $\dim(\partial W)$ via degeneracy log on the lowest K_α^W shell, with leading scaling $\dim(\partial W) \cdot \log n_{\max}$. The result is ring-free (a logarithm of an integer).*

The two operations act on the same modular state ρ_W but extract categorically different *continuum observables: the universal F-coefficient versus the wedge boundary dimension. The F-coefficient does* not *appear in $S(\rho_W)$.*

This decomposition is consistent with the PSLQ-disjointness finding of Sprint TD Track 5 (2026-05-09) [31] that GeoVac entropies generally live outside the master Mellin engine ring, while spectral-side observables generally live inside it.

4.4 Three blockers for literal Ryu–Takayanagi on the framework's graph

The continuum Ryu–Takayanagi formula [27] identifies the boundary entanglement entropy of a region A with the area of the bulk minimum surface anchored at ∂A, weighted by $1/(4G_N)$. A literal discrete analog on the framework's graph would require: (i) a well-defined notion of graph minimum-cut between spatial sub-regions; (ii) a perfect-tensor structure at graph vertices ensuring isometric bulk-boundary encoding; (iii) a bulk dual geometry.

Proposition 4.2 (Three blockers for literal RT on the framework)**.** *The framework's discrete S^3 graph at finite $n_{\max}$ exhibits three independent structural obstructions to a literal RT formula:*

1. *The Fock graph at level $n_{\max}$ has $\beta_0 = n_{\max}$ connected components (one per ℓ-sector), so the naive graph-cut between spatial sub-regions is identically zero or trivially confined to a single ℓ-sector.*

2. *The Wigner 3j symbols at graph vertices are Clebsch–Gordan projections rather than isometries; the HaPPY tensor-network RT proof [29] requires perfect tensors at network nodes.*

3. *No bulk-dual construction exists in the framework's infrastructure. The natural candidate — Wick continuation $S^3 \to H^3$ — was closed by Sprint RH-B (2026-04-17) [5] as a clean structural dead end: no discrete subgroup $\Gamma \subset SO(3,1)$ is selectable from framework invariants.*

These three blockers are not in tension with the boundary-side results of Sec. 3 or Sec. 5: the framework reproduces boundary observables on the same wedge KMS state, but cannot independently construct a bulk dual against which to compare via RT.

5 Casini–Huerta–Myers identification

5.1 Continuum CHM hemisphere modular Hamiltonian

The Casini–Huerta–Myers formula [25] for the continuum modular Hamiltonian of a hemisphere of S^3 in free CFT$_3$ is

$$K_{\mathrm{cont}} = 2\pi \int_{H_+} \xi(x) \cdot T^{00}(x)\, d\Omega(x), \tag{13}$$

where ξ is the wedge-preserving Killing vector field of the rotation that fixes the equator S^2_{equator}, and T^{00} is the local energy density. The flow generated by $K_{\mathrm{cont}}/(2\pi)$ is 2π-periodic, reflecting the canonical Bisognano–Wichmann period of the modular flow [26]; the eigenvalues of K_{cont} in the single-particle basis are $2\pi J$ for integer $J = 1, 2, 3, \ldots$.

5.2 Framework identification

The framework's wedge modular Hamiltonian (Paper 42 Def 5.1) is

$$K^W_\alpha = J_{\mathrm{polar}},$$

the rotation generator of the wedge-preserving $SO(2)$ on the truncated Camporesi–Higuchi spectral triple. Its spectrum is

$$\mathrm{spec}(K^W_\alpha) = \{2m_j : m_j = 1/2, 3/2, \ldots, (2n_{\max} - 1)/2\} = \{1, 3, 5, \ldots, 2n_{\max} - 1\},$$

integer-valued in the canonical $(2\pi)^{-1}$ rescaling, with multiplicity $g(2k+1) = (n_{\max} - k)(n_{\max} - k + 1)$. The modular flow $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$ closes bit-exactly at every tested $n_{\max}$ [9].

Theorem 5.1 (CHM identification at finite cutoff)**.** *The framework's modular Hamiltonian K^W_α at finite $n_{\max}$ is the operator-system-level analog of the continuum CHM modular Hamiltonian K_{cont} of (13), with:*

- *Both spectra integer-valued in canonical $(2\pi)^{-1}$ units.*
- *Both generate 2π-periodic modular flow.*
- *The framework's odd-integer spectrum $\{1, 3, \ldots, 2n_{\max} - 1\}$ corresponds to the half-integer Casimir grading of the wedge-preserving Killing field on the Weyl spinor sector.*
- *Convergence $K^W_\alpha \to K_{\mathrm{cont}}$ as $n_{\max} \to \infty$ holds in the Latrémolière propinquity at rate*

  $$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \le C_3 \cdot \gamma_{n_{\max}}$$

  with $C_3 \to 1$ and $\gamma_{n_{\max}} \sim (4/\pi)\log(n_{\max})/n_{\max}$ inherited from Paper 38 [8] via Paper 45 [12].

Proof sketch. The wedge-preserving Killing field of the round S^3 rotates the spinor harmonics by their m_j eigenvalue; the canonical 2π-period of the rotation gives the integer $2m_j$ spectrum in the canonical normalization. The framework's truncation $K^W_\alpha = J_{\mathrm{polar}}$ acts on the projected wedge sub-space $P_W \mathcal{H}^{n_{\max}}_{\mathrm{GV}}$ with exactly the same $2m_j$ eigenvalues, restricted to the multiplicities allowed by the truncation. Convergence in the propinquity follows from [8, 12]. □

6 Bulk-side scope

The three boundary-side observables (Sec. 3, Sec. 4, Sec. 5) all reproduce or identify their continuum CFT$_3$-on-S^3 analogs at finite cutoff. The corresponding *bulk-side* identifications — AdS$_4$ gravity, JLMS bulk-modular reconstruction, RT minimum surfaces, holographic entanglement entropy — are blocked at the framework's existing infrastructure.

Proposition 6.1 (Bulk side blocked)**.** *None of the bulk-side identifications corresponding to Sec. 3–5 are reachable from the framework's current infrastructure:*

1. *No AdS$_4$ / H^4 machinery exists in the framework's codebase (`geovac/`); a search for AdS, hyperbolic-4-space, or holographic primitives returns no matches.*

2. *Sprint RH-B (2026-04-17) [5] closed the natural candidate — Wick continuation $S^3 \to H^3$ — as a clean structural dead end: the continuation supplies no natural discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants.*

3. *Sprint L3e-P3 (2026-05-23) found that Paper 38's $4/\pi$ rate does* not *transport to the non-compact Coulomb setting, suggesting that any discrete-to-bulk convergence statement would require a non-compact rate analog that is itself an open NCG-math problem.*

The structural reading of Prop. 6.1 is that the bulk-side identification falls into Class 1 (calibration-external) of the framework's external-input partition [32]: the framework determines the boundary-side observables structurally but does not autonomously generate the bulk-dual geometry. Importing AdS_4 infrastructure would be a multi-month engineering commitment outside the present paper's scope.

7 Extension to S^5: structural refinement of the dual-basis decomposition

The bit-exact match of Sec. 3 extends to round S^5 via the same Hurwitz-zeta machinery. The Camporesi–Higuchi spectra on S^5 are spectrally analogous to S^3: the conformally coupled scalar has eigenvalues $(n+3/2)(n+5/2)$ with multiplicity $(2n+4)(n+1)(n+2)(n+3)/6$ for $n = 0, 1, 2, \ldots$, and the Weyl Camporesi–Higuchi Dirac has $|\lambda_n| = n + 5/2$ with multiplicity $(n+1)(n+2)(n+3)(n+4)/12$. The framework's spectral-zeta machinery on these spectra produces partition function values in a closed-form ring extending [21]'s S^3 ring by one $\zeta(5)/\pi^4$ generator.

7.1 Bit-exact closed forms on S^5

Theorem 7.1 (Conformally coupled scalar on S^5)**.** *The framework's scalar zeta-derivative on round unit S^5 satisfies*

$$F_s^{S^5} := -\tfrac{1}{2}\,\zeta'_{\Delta_{conf}}(0)^{S^5} = -\frac{\log 2}{32} - \frac{\zeta(3)}{32\pi^2} + \frac{15\,\zeta(5)}{64\pi^4} \approx -0.02297. \tag{14}$$

Proof sketch. The Hurwitz expansion

$$\zeta^{\mathrm{conf}}_{S^5}(s) = \frac{1}{3}\sum_{k\geq 0} g_k(s)[\zeta_R(2s+2k-4) - \zeta_R(2s+2k-2)] \tag{15}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ follows from the re-indexed multiplicity structure $v^2(v^2-1)/3$ at $v = n+2$. At $s = 0$, only $g_0(0) = 1$ survives, giving $\zeta^{\mathrm{conf}}_{S^5}(0) = 0$; the derivative

$$\zeta'_{S^5}(0) = \frac{2}{3}[\zeta'_R(-4) - \zeta'_R(-2)] + \frac{1}{3}\sum_{k\geq 1}\frac{\zeta_R(2k-4) - \zeta_R(2k-2)}{k \cdot 4^k} \tag{16}$$

converges geometrically in k. PSLQ at 200 dps and $k_{\max} = 100$ finds the integer relation $[32, -2, -2, 15]$ bit-exactly identifying

$$\zeta'_{S^5}(0) = \frac{\log 2}{16} + \frac{\zeta(3)}{16\pi^2} - \frac{15\,\zeta(5)}{32\pi^4}. \tag{17}$$

The standard identities $\zeta'_R(-2) = -\zeta(3)/(4\pi^2)$ and $\zeta'_R(-4) = +3\zeta(5)/(4\pi^4)$ deliver the $\zeta(5)/\pi^4$ contribution; the $\log 2$ piece arises from $\zeta'_R(0) = -\frac{1}{2}\log(2\pi)$ via the Hurwitz analytic-continuation tail. $\square$

Theorem 7.2 (Weyl Camporesi–Higuchi Dirac on S^5)**.** *The framework's Weyl Camporesi–Higuchi Dirac zeta-derivative on round unit S^5 satisfies, as an exact symbolic identity,*

$$D'_{Dirac}(0)^{S^5} = -\frac{3\log 2}{128} - \frac{5\,\zeta(3)}{128\pi^2} - \frac{15\,\zeta(5)}{256\pi^4} \approx -0.02163. \tag{18}$$

Proof sketch. The Dirichlet series

$$D_{\text{Dirac}}(s)^{S^5} = \frac{1}{12}\left[\zeta_H(s-4,5/2) - \tfrac{5}{2}\zeta_H(s-2,5/2) + \tfrac{9}{16}\zeta_H(s,5/2)\right] \tag{19}$$

follows from re-indexing $u = n + 5/2$ in the multiplicity $(n+1)(n+2)(n+3)(n+4)/12 = (u^2-9/4)(u^2-1/4)/12$. The half-integer Hurwitz function $\zeta_H(s,5/2)$ has closed form $(2^s-1)\zeta_R(s) - 2^s - (2/3)^s$. Differentiating each shift symbolically at $s=0$ and combining via the multiplicity coefficients $(1,-5/2,9/16)$ delivers the claimed result. □

7.2 Log 3 cancellation theorem

Theorem 7.3 ($\log 3$ cancellation on S^5)**.** *The $(2/3)^s$ terms in $\zeta_H(s-a,5/2)$ for $a \in \{0,2,4\}$ contribute, at $s=0$, factors of $\log(2/3) = \log 2 - \log 3$ scaled by $(2/3)^{-a} = (3/2)^a$. The coefficient of $\log 3$ in $D'_{Dirac}(0)^{S^5}$ evaluates to*

$$\frac{1}{12}\left[\frac{81}{16} - \frac{5}{2}\cdot\frac{9}{4} + \frac{9}{16}\cdot 1\right] = \frac{1}{12}\left[\frac{81-90+9}{16}\right] = 0 \tag{20}$$

exactly.

Remark 7.4. The structural combination $(1,-5/2,9/16)$ encoded in the multiplicity polynomial $(u^2-9/4)(u^2-1/4) = u^4-(5/2)u^2+9/16$ aligns the three shift operators $\{\zeta_H(s,5/2), \zeta_H(s-2,5/2), \zeta_H(s-4,5/2)\}$ such that all $\log 3$ contributions cancel exactly. The framework's spectral content on S^5 stays in the M-engine ring $\{\log 2, \zeta(3)/\pi^2, \zeta(5)/\pi^4\}$, even though the naive Hurwitz expansion suggests a contribution from $\log 3$. The same mechanism extends the M2/M3 ring statement from S^3 to S^5.

7.3 Dual-basis projection: structural-specificity to S^3

Theorem 3.3 of Sec. 3 establishes that on S^3, the (scalar, Dirac) pair admits a dual-basis projection onto the M2/M3 axes: $F_D + 2F_s = \log(2)/2$ (M2 only), $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ (M3 only). On S^5, the analogous statement *fails.*

Proposition 7.5 (Dual-basis non-extension to S^5)**.** *The Paper 50 Theorem 3.3 dual-basis projection does not extend to S^5. The M-engine ring on S^5 is three-dimensional ($\log 2$, $\zeta(3)/\pi^2$, $\zeta(5)/\pi^4$), but the (scalar, Dirac) plane is two-dimensional. The system $aF_s^{S^5} + bF_D^{S^5} = c\cdot$ (single M-axis) is over-determined: imposing isolation of one M-axis requires three linear conditions on two parameters (a,b), which has no nontrivial solution.*

Explicitly, attempting to isolate $\log 2$:

$$\zeta(3)/\pi^2 \textit{ coefficient} = 0 \implies -4a - 5b = 0 \implies a = -5b/4, \tag{21}$$

$$\zeta(5)/\pi^4 \textit{ coefficient} = 0 \implies 60a - 15b = 0 \implies a = b/4, \tag{22}$$

which are inconsistent unless $b = 0$.

Remark 7.6 (Conjecture for general odd d — partially falsified at S^7)**.** The structural-specificity of Theorem 3.3 to S^3 has a natural conjectural generalization. Round S^d for odd $d \geq 3$ is *conjectured* to have F in a ring of dimension $(d-1)/2+1$ generated by $\{\log 2, \zeta(3)/\pi^2, \zeta(5)/\pi^4, \ldots, \zeta(d-$

$2)/\pi^{d-3}\}$, and the dual-basis projection on free CFT species would require $(d-1)/2+1$ independent species: scalar plus $(d-1)/2-1$ higher-spin CFTs.

Status: The conjecture is verified bit-exactly at S^3 (Sec. 3) and S^5 (Theorems 7.1, 7.2). **It does not hold in this simple form at S^7.** The framework's value $\zeta'_{S^7,\mathrm{conf}}(0) \approx -0.00159466$ (computed to 30+ digits via the same Hurwitz machinery) does *not* admit a PSLQ integer relation in the 4-dimensional ring $\{\log 2, \zeta(3)/\pi^2, \zeta(5)/\pi^4, \zeta(7)/\pi^6\}$ at tolerance 10^{-30} to 10^{-80} with maxcoeff up to 10^{16}. Three structurally distinct alternative bases (log 2 basis plus $\{\log\pi\}$, $\{\log 3\}$, $\{\zeta(9)/\pi^8\}$, or $\{\log 2, \zeta(3), \zeta(5), \zeta(7), \pi^2, \pi^4, \pi^6\}$ with constants released) all return no relation. The structural pattern observed at S^3 and S^5 either requires an enlarged basis at S^7 (Dirichlet L-values, Catalan-class constants, or product transcendentals such as $\log 2 \cdot \zeta(3)/\pi^2$) or fails outright. The Hurwitz expansion is sound; the issue is structural, not computational.

Completing the dual basis on S^5 via a third free CFT species (the original motivation for the conjecture's extension) remains the natural next investigation. The S^7 negative result documents a clean boundary: the simple $\{\log 2, \zeta(\mathrm{odd})/\pi^{\mathrm{even}}\}$ ring is verified at $S^{3,5}$ and does not extend naively to S^7.

7.4 Implementation effort and continuation pattern

The S^5 extension was computed in 35 minutes of focused main-session work, continuing the discrete-tractability advantage documented across the math.OA arc. The `debug/ads_track_a_s5_{scalar, dirac}_partition_function.py` scripts implement the Hurwitz expansions and PSLQ verifications; `debug/ads_track_a_s5_extension_memo.md` documents the findings in full.

8 Synthesis and open questions

8.1 Unified picture

The framework's wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ supports three structurally distinct calculational operations, each extracting a different continuum CFT$_3$-on-S^3 boundary observable:

Operation	Continuum analog	Ring
$-\frac{1}{2}\zeta'_\Delta(0)$ (Theorem 3.1)	F-theorem coefficient	M2 + M3
$-\mathrm{Tr}[\rho_W \log \rho_W]$ (Prop. 4.1)	Wedge boundary dimension	ring-free
$\mathrm{spec}(K_\alpha^W)$ (Theorem 5.1)	CHM modular Hamiltonian	ring-free

All three converge to their continuum analogs as $n_{\max} \to \infty$ per [8, 12].

8.2 Open questions

1. **Other free-field CFTs.** The scalar and Dirac cases exhibit the orthogonal M2/M3 decomposition of Theorem 3.3. Does the same projection structure extend to higher-spin partition functions on S^3, in particular the Maxwell ($p=1$ form) and graviton ($p=2$) cases? The Maxwell partition function on round S^3 has known closed forms [23]; testing whether the framework's spectral-zeta machinery reproduces it (and whether the M2/M3 decomposition extends) is a natural follow-up.

2. **Squashed-S^3 generalization.** The KPS values above are for the round (unit) S^3. The squashed-S^3 partition functions of supersymmetric and free CFTs [22] use deformed half-integer Hurwitz machinery that may not reduce to the framework's MR-B Jacobi-ϑ_2 closed form. Whether the framework can match the squashed case is open.

3. **Propinquity-rate convergence statement.** Theorem 5.1 states convergence at the propinquity-bound level. A quantitative rate for the partition-function-side convergence $|F_{n_{\max}} - F^{\mathrm{KPS}}| \le C \cdot \gamma_{n_{\max}}$ with explicit C requires a Mellin-domain version of the Berezin-reconstruction framework of Paper 38; this is open work.

4. **Bulk-side identification (Prop. 6.1).** The JLMS bulk-modular Hamiltonian identification, the RT entanglement-entropy formula, and the AdS_4 dual geometry remain open. Importing AdS_4 / H^4 infrastructure is the named multi-month engineering commitment that would reach these.

8.3 Place in the GeoVac math.OA series

This paper is the twelfth math.OA standalone in the GeoVac series, following Papers 38, 39, 40, 42, 43, 44, 45, 46, 47, 48, 49. The previous eleven established operator-algebraic legitimacy of the framework's discrete-spectral-triple truncation, the propinquity convergence theory at various levels (Riemannian, tensor-product, universal compact-Lie-group, K^+-weak-form Lorentzian, strong-form Lorentzian, norm-resolvent non-compact, K^+-weak-form synthetic Lorentzian via Mondino–Sämann bridge, strong-form synthetic Lorentzian via OSLPLS). The present paper is the *first connecting that math.OA arc to a physics literature* — specifically the CFT-on-sphere partition function literature of [20–24] — with a load-bearing structural finding (the master Mellin engine verification on independent physics data) plus three bit-exact / formal correspondences (partition function, wedge boundary dimension, CHM modular Hamiltonian).

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References

[1] J. Loutey, *Paper 7: Dimensionless Vacuum*, GeoVac internal preprint, 2026.

[2] J. Loutey, *Paper 18: Exchange Constants and Transcendental Taxonomy*, GeoVac internal preprint, 2026.

[3] J. Loutey, *Paper 22: Angular Sparsity Theorem on S^3*, GeoVac internal preprint, 2026.

[4] J. Loutey, *Paper 23: Nuclear Shell Model on the Bargmann–Segal Lattice*, GeoVac internal preprint, 2026.

[5] J. Loutey, *Paper 29: Ramanujan Hopf Graphs and Wick-Rotation Scope*, GeoVac internal preprint, 2026.

[6] J. Loutey, *Paper 32: The GeoVac Spectral Triple*, GeoVac internal preprint, 2026.

[7] J. Loutey, *Paper 35: Time as Projection*, GeoVac internal preprint, 2026.

[8] J. Loutey, *Paper 38: SU(2) Propinquity Convergence on the Camporesi–Higuchi Spectral Triple*, GeoVac internal preprint, 2026.

[9] J. Loutey, *Paper 42: Tomita–Takesaki Modular Structure and Four-Witness Wick Rotation*, GeoVac internal preprint, 2026.

[10] J. Loutey, *Paper 43: Lorentzian Extension of the Four-Witness Theorem*, GeoVac internal preprint, 2026.

[11] J. Loutey, *Paper 44: Operator-System Substrate for Lorentzian Propinquity*, GeoVac internal preprint, 2026.

[12] J. Loutey, *Paper 45: K^+-Weak-Form Lorentzian Propinquity Convergence*, GeoVac internal preprint, 2026.

[13] J. Loutey, *Paper 46: Strong-Form Lorentzian Propinquity Convergence*, GeoVac internal preprint, 2026.

[14] J. Loutey, *Paper 47: Norm-Resolvent Convergence to the Non-Compact Lorentzian Krein Spectral Triple*, GeoVac internal preprint, 2026.

[15] J. Loutey, *Paper 48: Krein–Mondino–Sämann Bridge*, GeoVac internal preprint, 2026.

[16] J. Loutey, *Paper 49: OSLPLS Strong-Form Bridge and Twin Paradox as Quantum Information*, GeoVac internal preprint, 2026.

[17] V. Fock, *Zur Theorie des Wasserstoffatoms*, Z. Phys. **98** (1935), 145–154.

[18] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18.

[19] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067, arXiv:2004.14115.

[20] J. L. Cardy, *Operator content of two-dimensional conformally invariant theories*, Nucl. Phys. B **270** (1986), 186–204.

[21] I. R. Klebanov, S. S. Pufu, and B. R. Safdi, *F-theorem without supersymmetry*, JHEP **10** (2011) 038, arXiv:1105.4598.

[22] D. L. Jafferis, I. R. Klebanov, S. S. Pufu, and B. R. Safdi, *Towards the F-theorem: $N = 2$ field theories on the three-sphere*, JHEP **06** (2011) 102, arXiv:1103.1181.

[23] M. Beccaria and A. A. Tseytlin, *$\mathcal{N} = 4$ conformal supergravity: the complete one-loop dilatation operator*, JHEP **04** (2017) 100, arXiv:1702.02325.

[24] J. Lei and W. D. van Leuven, *Modularity in $d > 2$ free conformal field theory*, arXiv:2406.01567, 2024.

[25] H. Casini, M. Huerta, and R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, arXiv:1102.0440.

[26] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976), 303–321.

[27] S. Ryu and T. Takayanagi, *Holographic derivation of entanglement entropy from AdS/CFT*, Phys. Rev. Lett. **96** (2006), 181602, arXiv:hep-th/0603001.

[28] D. L. Jafferis, A. Lewkowycz, J. Maldacena, and S. J. Suh, *Relative entropy equals bulk relative entropy*, JHEP **06** (2016) 004, arXiv:1512.06431.

[29] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, *Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence*, JHEP **06** (2015) 149, arXiv:1503.06237.

[30] GeoVac internal, *Sprint BH-Phase0 diagnostic memo*, `debug/bh_phase0_diagnostic_memo.md`, 2026-05-22.

[31] GeoVac internal, *Sprint TD Track 5 PSLQ-disjoint correlation entropy*, `debug/sprint_td_track5_memo.md`, 2026-05-09.

[32] GeoVac internal, *External-input three-class partition*, `memory/external_input_three_class_partition.m`
2026-05-25.